MAT 226 Linear Algebra

Muhammad Abdullah

Homework # 8

Exercise 3.2

6) $\begin{bmatrix} 1 & 5 & -3 \\ 3 & -3 & 3 \\ 2 & 13 & -7 \end{bmatrix}$

$R_2 \leftarrow R_2 - 3R_1$
$R_3 \leftarrow R_3 - 2R_1$

$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -18 & 12 \\ 0 & 3 & -1 \end{bmatrix}$

$R_3 \leftrightarrow R_2$ (multiply det by -1)

$\begin{bmatrix} 1 & 5 & -3 \\ 0 & 3 & -1 \\ 0 & -18 & 12 \end{bmatrix}$

$R_3 \leftarrow R_3 + 6R_2$

$\begin{bmatrix} 1 & 5 & -3 \\ 0 & 3 & -1 \\ 0 & 0 & 6 \end{bmatrix}$

determinant $= -1 \times 1 \times 3 \times 6$
$= -18$

12) $\begin{bmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 5 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{bmatrix}$

$R_3 \leftarrow R_3 - 2R_4$

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ -3 & 0 & -2 & 0 \\ 4 & 2 & 4 & 3 \end{bmatrix}$$

non zero term at $(4,4)$

$(-1)^{4+4} = 1$

cofactor expansion down column 4

$$\det = 3\begin{bmatrix} -1 & 2 & 3 \\ 3 & 4 & 3 \\ -3 & 0 & -2 \end{bmatrix}$$

$R_2 \leftarrow R_2 - 2R_1$

$$\det = 3\begin{bmatrix} -1 & 2 & 3 \\ 5 & 0 & -3 \\ -3 & 0 & -2 \end{bmatrix}$$

cofactor expansion down column 2

non zero at $(1,2)$ so $(-1)^3 = -1$

$$\det = 3 \times -2 \times \begin{bmatrix} 5 & -3 \\ -3 & -2 \end{bmatrix}$$

$$= -6\begin{bmatrix} 5 & -3 \\ -3 & -2 \end{bmatrix}$$

$$= -6\left[(5\times -2) - (3\times -3)\right] = -6(-10-9) = +114$$

28) (a) True

single row interchange multiplies the determinant by -1. Two consecutive ones mean $(-1)(-1) = +1$ multiplying by $+1$ leaves the value of determinant unchanged.

(b) False

This not generally true and only applies if the matrix is triangular.

(c) False

counter example

$$\begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} \quad \det = 3\times12 - 6\times6 = 0$$

Determinant can also be zero when columns or rows form a linearly dependent set.

(d) False

$$\det A^T = \det A$$

The determinant of A^T is the exact same value as the determinant of A.

32) let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $n=2$ here

$$\det A = (1\times4 - 2\times3)$$

$$= -2$$

now

$$rA = r\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} r & 2r \\ 3r & 4r \end{bmatrix}$$

$$\det(rA) = 4r^2 - 6r^2$$
$$= -2r^2$$

since square term comes from $n = 2$

$$\det(rA) = r^n \det(A)$$

34) show

$$\det(PAP^{-1}) = \det(A)$$

$$\det(PAP^{-1}) = \det(P) \cdot \det(A) \cdot \det(P^{-1})$$
$$= \det(P) \cdot \det(A) \cdot \frac{1}{\det(P)}$$
$$= \det(A)$$

determinants are just scalar numbers and we can apply commutative law here.

Exercise 3.3

(2) $4x_1 + x_2 = 6$
$5x_1 + 2x_2 = 7$

$$x_i = \frac{|A_i|}{|A|} \qquad \begin{bmatrix} 4 & 1 \\ 5 & 2 \end{bmatrix}$$

$\det(A) = 4\times2 - 5\times1$

$= 8 - 5$

$= 3$

$A_1 = \begin{bmatrix} 6 & 1 \\ 7 & 2 \end{bmatrix}$ $\quad A_2 = \begin{bmatrix} 4 & 6 \\ 5 & 7 \end{bmatrix}$

$\det(A_1) = 6\times2 - 7\times1$

$= 12 - 7$

$= 5$

$\det(A_2) = 4\times7 - 6\times5$

$= 28 - 30$

$= -2$

$x_1 = \frac{|A_1|}{|A|} = \frac{5}{3}$ $\quad x_2 = \frac{|A_2|}{|A|} = \frac{-2}{3}$

20) $(0,0)$ $(-1,3)$ $(4,-5)$ $(3,-2)$

$\vec{u} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ & $\vec{v} = \begin{bmatrix} 4 \\ -5 \end{bmatrix}$

check $\begin{bmatrix} -1 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ -5 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ ✓✓

$A = \begin{bmatrix} -1 & 4 \\ 3 & -5 \end{bmatrix}$

$\det(A) = (-1\times-5) - (3\times4)$

$= 5 - 12$

$= -7$

area $= |-7| = 7$

Exercise 4.1

6) determine if set of all polynomials of form $p(t) = a + t^2$ is a subspace of $\mathbb{P}_2$

① check if it contains zero vector.

for it to be a zero polynomial, ($p(t) = 0$), all of its coefficients must be 0.

Here, t^2 is fixed at t^2 (and not at 0)
It is impossible to choose a real number a that turns the entire polynomial into 0.

It doesnot contain the zero vector, hence this set is <u>not</u> a subplace of $\mathbb{P}_2$

12)
$$\begin{bmatrix} 2s+4t \\ 2s \\ 2s-3t \\ 5t \end{bmatrix} = \begin{bmatrix} 2s \\ 2s \\ 2s \\ 0 \end{bmatrix} + \begin{bmatrix} 4t \\ 0 \\ -3t \\ 0 \end{bmatrix} = s\begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix} + t\begin{bmatrix} 4 \\ 0 \\ -3 \\ 5 \end{bmatrix}$$

W is a set of all linear combinations of the two specific vectors in $\mathbb{R}^4$.

Since W is the span of a set of vectors in $\mathbb{R}^4$; it is a subspace of $\mathbb{R}^4$.